

1. (Question 1) The Jupyter notebook file *Q1.ipynb* has all relevant code for this problem. Codex GPT-6 Astra assisted me in writing the code. After implementing a basic euclidean distance function, I created a K-nearest neighbors classifier that used the provided training file to predict on the test file for $k = 1 \dots 9$. Note that in the definition of the knn_classifier, ties are decided by choosing the label of the nearest neighbor since the labels array is sorted by increasing distance from least to greatest, and the max would return the first label in the case of a tie.

```
def euclidean_distance(dimension, point1, point2):
    squared_distance = 0
    for i in range(dimension):
        squared_distance += (point1[i] - point2[i]) ** 2
    return squared_distance ** 0.5

def knn_classifier(k, training_data, test_point):
    distances = []
    for data_point in training_data:
        distance = euclidean_distance(len(test_point), data_point
[: -1], test_point)
        distances.append((distance, data_point[-1]))
    distances.sort(key=lambda x: x[0])
    k_nearest_neighbors = distances[:k]
    labels = [neighbor[1] for neighbor in k_nearest_neighbors]
    return max(labels, key=labels.count)
```

I got the following results, which are also in the notebook file:

kNN results on the original, unscaled data

K	1	2	3	4	5	6	7	8	9
Correct predictions	462	462	462	462	462	462	462	462	462
Accuracy	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000

I was surprised that the accuracy was 100% for all k -values and I was concerned that I had implemented the algorithm incorrectly. But after reviewing my implementation, I think it is correct. Next, I implemented min-max normalization and standardization to scale the data and then re-ran my KNN algorithm. In both cases, if a certain feature in the training data had zero variance or was constant, I set that feature to 0.

```

def min_max_normalize(data, feature_ranges=None):
    # When no ranges are supplied, calculate them from the training data.
    if feature_ranges is None:
        feature_ranges = []
        for i in range(len(data[0]) - 1): # Exclude the final label column.
            column = [row[i] for row in data]
            feature_ranges.append((min(column), max(column)))

    normalized_data = []
    for row in data:
        normalized_row = []
        for i in range(len(row) - 1):
            min_value, max_value = feature_ranges[i]
            if max_value == min_value:
                normalized_value = 0.0 # Ignore a constant training feature
            else:
                normalized_value = (row[i] - min_value) / (max_value -
min_value)
            normalized_row.append(normalized_value)
        normalized_row.append(row[-1]) # Keep the original class label.
        normalized_data.append(normalized_row)

    return normalized_data, feature_ranges

```

```

def standardize(data, feature_stats=None):
    # When no stats are supplied, calculate them from the training data.
    if feature_stats is None:
        feature_stats = []
        for i in range(len(data[0]) - 1): # Exclude the final label column.
            column = [row[i] for row in data]
            mean = sum(column) / len(column)
            variance = sum((x - mean) ** 2 for x in column) / len(column)
            std_dev = variance ** 0.5
            feature_stats.append((mean, std_dev))

    standardized_data = []
    for row in data:
        standardized_row = []
        for i in range(len(row) - 1):
            mean, std_dev = feature_stats[i]
            if std_dev == 0:
                standardized_value = 0.0 # Ignore a constant training
feature.
            else:
                standardized_value = (row[i] - mean) / std_dev
            standardized_row.append(standardized_value)
        standardized_row.append(row[-1]) # Keep the original class label.
        standardized_data.append(standardized_row)

    return standardized_data, feature_stats

```

I re-ran the KNN classifier on the two versions of the scaled data and got the following results:

Min-max normalization									
K	1	2	3	4	5	6	7	8	9
Correct predictions	461	461	461	461	461	461	461	461	461
Accuracy	0.998	0.998	0.998	0.998	0.998	0.998	0.998	0.998	0.998

Standardization									
K	1	2	3	4	5	6	7	8	9
Correct predictions	461	461	461	461	461	461	461	461	461
Accuracy	0.998	0.998	0.998	0.998	0.998	0.998	0.998	0.998	0.998

Both scaling methods resulted in a slight decrease in accuracy, misclassifying only one test point. I was again suprised because I had expected that scaling would only improve accuracy, or at least not hurt it. Given that the original data had perfect accuracy, I thought the scaled versions would also. I was again worried that I had implemented the algorithm incorrectly, but I cannot identify what is wrong with the implementation so I am working on the assumption that the given data has been set up this way on purpose and that my implementation is correct.

Given these results, it would appear that the original unscaled data is the best approach for this particular dataset, but that is not necessarily true in general. I wanted to know which point was misclassified, so the additional code in the file was created by Codex to help me identify which point label was changed after scaling.

In general, I would expect scaling to improve accuracy for the KNN algorithm so that no one feature has an outsized influence on the calcuated distance, but if one feature already has low variance, it's change from the original data to scaling might mean that after scaling it can influence the classification. Feature 4 from the csv appears to do that in this case. I became curious about whether we could drop a feature entirely and still perfectly predict the test set. I prompted Codex to check all possible subsets of the features and run the KNN algorithm for the original data and the two different scaled data. This code showed that if we dropped Feature 4 and used the standardized data, we could stil get 100% accuracy for $k = 1...9$. I think this supports my hypothesis that Feature 4 is the one that is causing the misclassification after scaling.

2. (*Question 2*) These summaries are written by me. I asked for feedback from Codex on what I wrote and made some edits based on AI feedback, but I am the author.

The first paper that I read was *Deep Nearest Neighbor Anomaly Detection* by Bergman et al. (2020). The paper proposes and tests using an available neural network image classifier to first extract features of the image and then perform kNN on the extracted features to detect anomalous images. They call it Deep Nearest Neighbors (DN2). Identifying anomalous images is advantageous in threat assessment and preventative maintenance. For example, a stream monitoring an oil pipeline might be able to detect evidence of corrosion and imminent failure. Anomaly detection training can be *supervised*, *semi-supervised*, or *unphunsupervised*. In particular, they were interested in the *semi-supervised* cases where the training set has no anomalous images and *unsupervised* cases where unlabeled normal and

anomalous images are in the training set. The authors ran the ImageNet pre-trained ResNet feature extractor on both the training and test images. The authors note that the majority of training time comes from neural net inference to extract features and that more layers produces better results, but that DN2 is able to perform well even with a very small training set, as evidenced by their figure showing average ROC area. They also investigated identifying groups of images rather than single images, but noted that larger groups makes the task more difficult. They also noted that kNN time increases with larger datasets, but they suggested first doing k-means and then performing kNN on the cluster centers to reduce the number of comparisons. Besides time to classification increasing with large data sets, the authors noted that the pretrained features might be less effective on images very dissimilar to those in ImageNet.

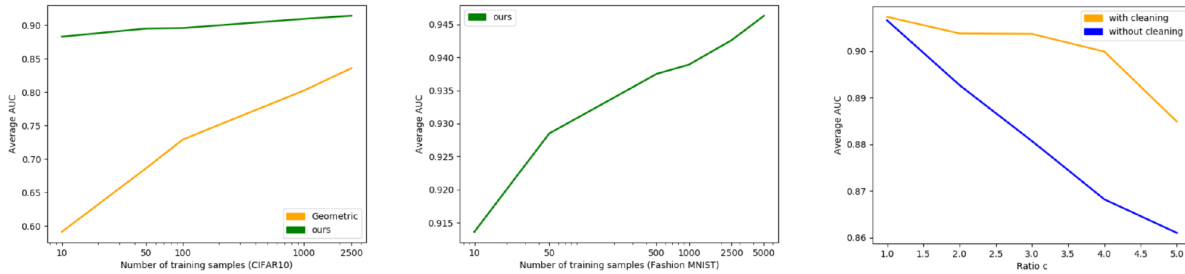


Figure 1: Reproduced from Figure 5 of Bergman et al. (2020).

This idea of reducing the size of the training set in particular stood out to me. Cluster centers make sense as a representative reduction of a larger set, but I am curious if we not just place the rough location of a test point and then just test the values within a ball of a given radius of that test point rather than comparing to the entire training corpus. Perhaps an algorithm that could quickly place the test point in an approximate “quadrant” of available training data and then only compare to the training points in that quadrant. If the training data was already processed, not unlike k-means clustering, then we could already know a subset of training data points that are near a given training data point. Specifying the algorithm for grouping the training points and comparing it’s speed and accuracy to k-means clustering and then kNN on the cluster centers would be one way to test this idea. Codex tells me that my idea is close to the kd-trees mentioned in the paper.

The second paper I read was *Deep k-Nearest Neighbors: Towards Confident, Interpretable and Robust Deep Learning* by Papernot and McDaniel (2018). While the previous paper looked at using a neural network to first process images that could then be analyzed with k nearest neighbors, this paper attempts to improve neural network performance with the kNN algorithm. In particular, the authors were interested in looking at the interpretability of a deep neural network at intermediate stages as well as improving the robustness of DNN models against adversarial attacks. Their idea was to perform kNN on each intermediate layer output to check how similar that output was to expected outputs for the training data. As they say in the paper, they wanted to answer “Why did the model decide that?”. In the included figure from the paper, you can see how they are using k nearest neighbors to check intermediate outputs from the neural network and see if they are conformal, that is, the match all through through the network during the hidden layers.

Their results show that by checking intermediate outputs from the layers of the neural network and comparing them with kNN, their DkNN model has a higher accuracy because of lower credibility (it’s estimate of the probability of correct classification) than traditional DNNs and is also more robust against adversarial attacks because of this lower credibility. They were even able to find mislabeled data in the MNIST data set by searching for inputs with high credibility in the model but had been marked as incorrect predictions. The models also outperform softmax on out of training examples because they had low credibility on the new data. The authors seem to think this approach is very strong, as the scenario they describe for a successful adversarial attack requires inside knowledge and would mean targeting an image that stays conformal within the intermediate stages of the DkNN despite ultimately being an anomalous image.

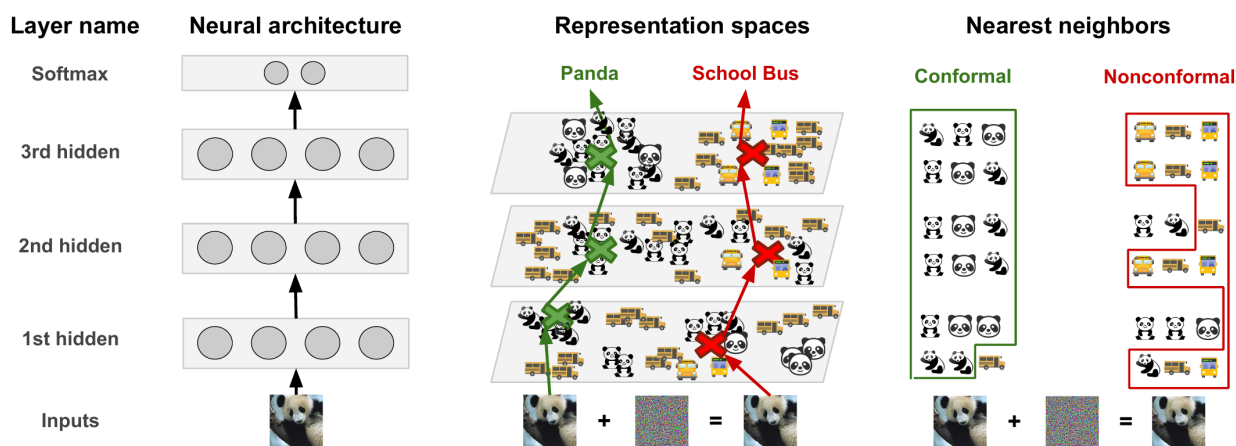


Fig. 1: **Intuition behind the Deep k-Nearest Neighbors (DkNN)**—Consider a deep neural network (left), representations output by each layer (middle) and the nearest neighbors found at each layer in the training data (right). Drawings of pandas and school buses indicate training points. Confidence is high when there is homogeneity among the nearest neighbors labels (e.g., here for the unmodified panda image). Interpretability of the outcome of each layer is provided by the nearest neighbors. Robustness stems from detecting nonconformal predictions from nearest neighbor labels found for out-of-distribution inputs (e.g., an adversarial panda) across different layers. Representation spaces are high-dimensional but depicted in 2D for clarity.

Figure 2: Reproduced from Figure 1 of Papernot and McDaniel (2018), including the original caption.

3. (Question 3)

- The prior probabilities are:

$$P(\text{Yes}) = \frac{\text{Number of stolen vehicles}}{\text{Total number of vehicles}} = \frac{5}{10} = 0.500,$$
$$P(\text{No}) = \frac{\text{Number of non-stolen vehicles}}{\text{Total number of vehicles}} = \frac{5}{10} = 0.500.$$

- Using the Naive-Bayes assumption, the conditional likelihoods are:

$$P(\text{Red, Domestic, SUV} \mid \text{Yes}) = P(\text{Red} \mid \text{Yes}) P(\text{Domestic} \mid \text{Yes}) P(\text{SUV} \mid \text{Yes})$$
$$= \frac{3}{5} \cdot \frac{2}{5} \cdot \frac{1}{5} = 0.048.$$

$$P(\text{Red, Domestic, SUV} \mid \text{No}) = P(\text{Red} \mid \text{No}) P(\text{Domestic} \mid \text{No}) P(\text{SUV} \mid \text{No})$$
$$= \frac{2}{5} \cdot \frac{3}{5} \cdot \frac{3}{5} = 0.144.$$

- The unnormalized posterior scores are:

$$\text{Score}(\text{Yes}) = P(\text{Yes}) P(\text{Red, Domestic, SUV} \mid \text{Yes})$$
$$= 0.500 \cdot 0.048 = 0.024,$$
$$\text{Score}(\text{No}) = P(\text{No}) P(\text{Red, Domestic, SUV} \mid \text{No})$$
$$= 0.500 \cdot 0.144 = 0.072.$$

Since $0.072 > 0.024$, the predicted class is No (not stolen).

- If a required conditional count were zero, the entire class score would become zero. If we fix this by applying Laplace smoothing, we would add 1 to every possible value's count, including nonzero counts, and then add the number of possible values of that feature to the denominator. For example, since Type can be Sports or SUV, if the probability for Sports given Yes stolen was $\frac{0}{5}$ we would change the conditional likelihood to $\frac{0+1}{5+2} = \frac{1}{7}$. The conditional likelihoods for Color and Origin would also be shifted similarly.

4. (Question 4) Codex assisted me with this problem. I first began by investigating the data. Since there are no headers, I asked Codex to write some code that would give an enumeration of values or at least summary statistics in each category. Although quite a few features turned out to be binary, there were several that were not. After reviewing the models available from scikit, it seemed to me that since not all of the features were binary or clearly all the same kind of feature, I could not use one of the specialized models that made restrictive assumptions (like BernoulliNB for binary features). Since GaussianNB seems the most general and therefore appropriate for the data, I ran that, but Codex suggested we could also look at Categorical after binning the data (and possibly losing numeric features). I let Coded run both after the Gaussian just to see if there was any improvement.

Results from the code are reproduced in the tables below. For 5 misclassified points, I was unsure where to start to identify why Bayes might have misclassified them. Given that so many features are binary I thought that perhaps the discrete nature of a binary category means it is easy for a small difference to flip the indicator. I notice that Feature 10 for nearly all the misclassified points has values in the 30s. Perhaps looking at all points with those values and seeing how many are misclassified would produce some insights. I would also be interested in running kNN on the misclassified points just to compare them to their neighbors.

GaussianNB results. The scikit Gaussian implementation in Q4 . ipynb produced the following results.

Dataset	Rows	Input features	Total columns
Training	700	24	25
Test	300	24	25

Test metric	Value
Accuracy	0.740
Macro-precision	0.701
Macro-recall	0.714
Macro-F1	0.706

Actual class	Predicted class 1	Predicted class 2
Class 1	162	45
Class 2	33	60

Misclassified test examples.

Test row	Actual	Predicted	Encoded features (F1-F24)
1	2	1	4, 12, 2, 11, 3, 3, 2, 4, 3, 29, 3, 1, 1, 1, 1, 0, 0, 1, 0, 1, 0, 0, 1, 0
2	2	1	1, 48, 4, 63, 1, 5, 3, 4, 4, 46, 3, 2, 1, 2, 1, 0, 1, 1, 0, 0, 0, 0, 0, 1
5	1	2	2, 27, 2, 25, 1, 2, 2, 1, 2, 32, 3, 1, 2, 2, 1, 0, 0, 1, 0, 0, 1, 0, 0, 1
21	2	1	3, 9, 0, 13, 1, 2, 3, 2, 3, 34, 3, 2, 1, 2, 1, 0, 0, 1, 0, 0, 1, 0, 0, 0
31	1	2	2, 24, 3, 64, 1, 2, 3, 2, 3, 33, 3, 1, 1, 1, 1, 0, 0, 1, 0, 0, 1, 0, 0, 1

5. (Question 5) Solutions to this problem were created with Codex GPT-6 Astra. A log of my conversation was maintained by Astra in the Markdown file labeled q5_ai_conversation_log.md. Most of my prompts are recorded verbatim, but some longer ones have been summarized. I prompted Astra to write summaries of the work we did for this problem, but the reflection requested in Bullet Point 4 was written by me.

Identified problems and corrections. The corrected notebook is `q5_corrected.ipynb`. Line numbers refer to the original `q5_starter_classifier.py`. The explanations and notebook evidence below were prepared with AI assistance; independent validation is recorded separately in the AI-use conversation log.

- a. **CSV headers are parsed as numeric data (line 27).** The original `np.loadtxt` call does not skip the header row `F1, F2, F3, F4, Class`. It therefore fails while converting `F1` to a floating-point number, before any model can run. The correction skips the single header row:

```
np.loadtxt(path, delimiter=",", skiprows=1)
```

Evidence: Figure 3 shows the original loading failure. The corrected notebook successfully loads 910 training rows and 462 test rows, each with five numeric columns. Thus the correction resolves an execution failure rather than merely changing formatting.

- b. **The distance function sums signed differences (lines 30–31).** The expression `np.sum(a - b)` is not Euclidean distance: differences can cancel, and the result can be negative. In addition, $\sum_j (q_j - r_j) = \sum_j q_j - \sum_j r_j$, so sorting these scores favors training rows with larger coordinate sums independently of the query, apart from numerical tie effects. The corrected function computes the square root of the sum of squared coordinate differences:

```
return np.sqrt(np.sum((a - b) ** 2))
```

Evidence: For $(0, 0)$ and $(3, 4)$, the original result is -7.000 , whereas the corrected distance is 5.000 . For $(0, 0)$ and $(3, -3)$, the original result is 0.000 , whereas the corrected result is $\sqrt{18} \approx 4.243$. These examples demonstrate both a failure of the original formula and agreement of the correction with Euclidean geometry.

- c. **Class values are included in feature coordinates (lines 50–54).** The starter scales all five columns into `train_x` and `test_x`, including the final class column. Extracting labels afterward does not remove that column from the feature arrays. This exposes the true test labels to the predictor and makes the feature representation invalid for evaluating classification. The correction separates features and labels before preprocessing:

```
X_train = train[:, :-1]
y_train = train[:, -1].astype(int)
X_test = test[:, :-1]
y_test = test[:, -1].astype(int)
```

Only the four-column feature arrays are scaled and passed to the distance calculation; labels remain separate for voting and evaluation. **Evidence:** The original transformations preserve all five columns. The corrected notebook reports feature shapes $(910, 4)$ and $(462, 4)$. The slices explicitly exclude the last column while preserving its values in the separate label arrays.

- d. **Scaling is fitted using test data (line 50).** The starter combines training and test rows before fitting its scaler, so test values influence the means and standard deviations used to prepare the model inputs. This is preprocessing leakage and compromises the held-out evaluation, whether or not it increases measured accuracy. The correction fits on training features alone and reuses those parameters:

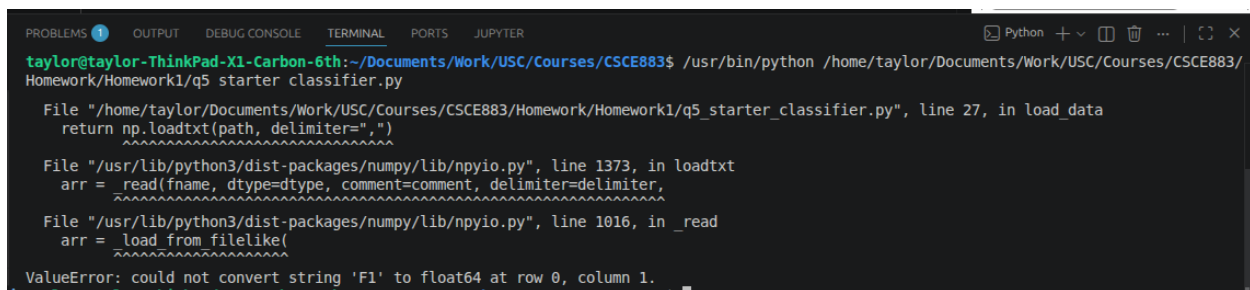
```
scaler = StandardScaler().fit(X_train)
train_scaled = scaler.transform(X_train)
test_scaled = scaler.transform(X_test)
```

The internal training/validation split likewise fits its scaler only on the fitting subset. **Evidence:** The pooled F2 mean is 1.922, while the training-only F2 mean is 2.010, demonstrating that test data alter the original fitted parameters. The corrected notebook verifies that the scaler's mean vector equals the training-feature mean vector.

- e. **Vote ties silently favor class 0 (line 38).** The starter counts votes by class and selects the index with the largest count using `argmax`. For equal binary class counts, `argmax` selects index 0. The corrected notebook instead adopts Question 1's stated policy: choose the label of the nearest neighbor whose class is tied for the most votes. This is a voting-policy alignment with Q1; Q5 does not independently prescribe a particular tie policy.

```
classes, counts = np.unique(votes, return_counts=True)
tied = classes[counts == counts.max()]
return next(label for label in votes if label in tied)
```

Here `votes` is ordered by increasing neighbor distance. **Evidence:** Nearest-first votes `[1, 0, 1, 0]` yield counts `[2, 2]`, so the starter chooses 0 while the adopted policy chooses 1. The notebook's four-neighbor toy example confirms that a tied vote returns the nearest neighbor's class, 1.



```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS JUPYTER
taylor@taylor-ThinkPad-X1-Carbon-6th:~/Documents/Work/USC/Courses/CSCE883$ /usr/bin/python /home/taylor/Documents/Work/USC/Courses/CSCE883/Homework/Homework1/q5 starter classifier.py
File "/home/taylor/Documents/Work/USC/Courses/CSCE883/Homework/Homework1/q5_starter_classifier.py", line 27, in load_data
    return np.loadtxt(path, delimiter=",")
File "/usr/lib/python3/dist-packages/numpy/lib/npio.py", line 1373, in loadtxt
    arr = read(fname, dtype=dtype, comment=comment, delimiter=delimiter,
File "/usr/lib/python3/dist-packages/numpy/lib/npio.py", line 1016, in _read
    arr = load from filelike(
ValueError: could not convert string 'F1' to float64 at row 0, column 1.
```

Figure 3: Q5 starter CSV-loading error: line 27 attempts to convert the column header F1 to a floating-point number.

Corrected model results. Both models use the same 462 test examples from `Q1_test.csv`, with standardization fitted only on training features. KNN uses $K = 1$, selected from $K = 1, \dots, 9$ using a stratified validation split within the training set. All nine values tied on validation macro-F1; the stated selection rule favored the smallest K . Gaussian Naive Bayes estimates a Gaussian density for each feature within each class and uses training class frequencies as priors. Its Gaussian and conditional-independence assumptions are approximations for the four numeric features.

Model	Accuracy	Macro-F1
Corrected KNN ($K = 1$)	0.998	0.998
Gaussian Naive Bayes	0.874	0.871

Confusion matrices below use rows for true classes and columns for predicted classes. Entries are counts; macro-F1 averages the F1 scores of classes 0 and 1 equally.

Corrected KNN ($K = 1$)			Gaussian Naive Bayes		
	Predicted 0	Predicted 1		Predicted 0	Predicted 1
True 0	261	1	True 0	238	24
True 1	0	200	True 1	34	166

Results are reproduced from the executed notebook `q5_corrected.ipynb` and its saved output `q5_corrected_output.txt`.

Reflection. Repeating the same data as Q1 helped with this problem because I already had an idea what to expect. My python syntax knowledge is a bit sparse, especially for syntactic sugar on numpy functions, but I was able to notice the basic issues with the wrong metric, including the class label in the point, and setting parameters using test values. The other issues Codex pointed out to me, although I feel as though I should have attempted to run the code before asking it to analyze, just like I read the code before asking Codex to analyze it.

The code does attempt to normalize the data and handle a 0 variance case. As pointed out in the responses to the earlier bullet points, there were issues with attempting to read the header, including the class label, using the test points in parameter training, and not explicitly deciding what to do in the case of a tie among neighbor votes. Looking at the results, we see that kNN does better than Naive Bayes. The additional investigation I asked Codex to conduct to see if not making the naive assumption about independence among features would improve the result gave the expected result. Between Naive Bayes and kNN I would recommend using kNN given how much better it performs on accuracy and F1. Improving Bayes by investigating dependence between features is not something I would want to try if I did not have Codex to work through that code for me to test it.

As stated before, Codex GPT-6 Astra assisted me with this problem. Codex kept a log of our conversation and work on Problem 5 and that log can be viewed in the included Markdown file named `q5_ai_conversation_log.md`.

References

- Liron Bergman, Niv Cohen, and Yedid Hoshen. Deep nearest neighbor anomaly detection. *arXiv preprint arXiv:2002.10445*, 2020. URL <https://arxiv.org/abs/2002.10445>.
- Nicolas Papernot and Patrick McDaniel. Deep k-nearest neighbors: Towards confident, interpretable and robust deep learning. *arXiv preprint arXiv:1803.04765*, 2018. URL <https://arxiv.org/abs/1803.04765>.